PROJECT REPORT

Hotel Booking & Occupancy Manager

A Java-based application for booking cost calculation and room occupancy management

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Project area	Java Programming / Hotel Booking Management

1. Introduction

The **Hotel Booking & Occupancy Manager** is a Java-based application that organizes booking details, calculates costs, tracks room availability, and categorizes customers by booking amount.

2. Problem Statement

Manual booking calculations and separate room records can lead to delays, errors, and booking conflicts. A computerized system helps make booking management more accurate and organized.

3. Objectives and Main Features

• Collect customer and booking details.
• Allow users to select a room type and enter the number of rooms and days.
• Calculate the total booking cost.
• Check membership categories.
• Track room occupancy and availability.

Feature	Purpose
Customer name and room type	Records the basic booking information.
Cost calculation	Calculates the stay cost from room price, rooms, and days.
Membership category	Groups customers according to the booking amount.
Occupancy tracking	Helps monitor room availability and usage.

4. Booking Cost and Membership

Total Booking Amount = Room Price × Number of Rooms × Number of Days

Example: ■2,000 × 2 rooms × 3 days = ■12,000.

The presentation lists Silver, Gold, and Platinum membership categories based on the total booking amount. The exact thresholds should follow the values shown in the project slides.

5. Program Flow

Start → Enter customer details → Select room type → Enter rooms and days → Calculate total amount → Check membership category → Check occupancy and availability → Display result.

6. Java Concepts Used

The project uses variables, data types, operators, if-else-if and switch statements, loops, methods, arrays, Scanner for input, and basic object-oriented programming.

7. Advantages

- Faster booking calculations
- Fewer manual errors
- Organized customer and booking details
- Easier room occupancy tracking
- Simple implementation using Java

8. Future Scope

Future improvements could include saving records permanently, searching and updating bookings, improving room-status management, and adding a graphical user interface.

9. Conclusion

The Hotel Booking & Occupancy Manager demonstrates how Java can support basic hotel booking tasks. By combining cost calculation, customer details, membership checks, and occupancy tracking, it aims to make booking management clearer and more efficient.

Reference: Project presentation, “Hotel Booking & Occupancy Manager,” prepared by Harini and Akshaya.